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Google Research introduces JAX library for computer vision research

With new architectures like vision transformers (ViTs) taking up day-to-day applications, there is a clear demand for software and machine learning



infrastructure to support easy and extensible neural network architecture research in the field of vision.

Researchers from Google Brain have introduced SCENIC, an open source JAX library with a focus on transformer

based models for computer vision research. It has been successfully used to develop classification, segmentation, and detection models for images, videos, and other modalities, including multi-modal setups.

The SCENIC toolkit aims to facilitate rapid experimentation, prototyping, and research of new vision architectures and models. It offers optimised implementations of state-of-the-art research models spanning a wide range of modalities.

This open source library offers a unified, all-in-one codebase for modelling needs and implementations like ViT, DETR, MLP Mixer, ResNet and U-Net.

SCENIC is developed in JAX and uses Flax as the neural network library. JAX is a simple-to-use library that allows automatic differentiation of native Python and NumPy functions. It can support multi-host and multi-device training on accelerators such as GPUs and TPUs, making it perfect for large scale machine learning research.

DIA integrates its Oracle infrastructure with the NEAR Protocol

DIA Association has announced the integration of its Oracle infrastructure with the layer 1 distributed ledger NEAR Protocol. This deployment is set to enable



the development of new decentralised applications with easy access to accurate and transparent data provided by DIA.

The NEAR Protocol, live on Mainnet since April 2020, is a blockchain network created and

developed by the NEAR Foundation, using a proof-of-stake consensus mechanism. The NEAR ecosystem encompasses several verticals including DeFi, with dApps like Maker; utility/infrastructure, with tools like The Graph; or games such as Near Lands. The protocol is also home to NFTs, enabling users to create, mint and sell non-fungible tokens through its engine Mintbase.

The blockchain ecosystem has seen a number of new efficiency driven chains surface, aiming to solve the scalability issue that ledgers like Ethereum are showing due to the increasing block sizes and number of daily transactions. DIA has

NASA and ESA release open source tool for analysing Earth science data

NASA and ESA (European Space Agency) have publicly released an open source science tool for analysing Earth science data in the cloud – the Multi-Mission Algorithm and Analysis Platform (MAAP).

MAAP enables scientists to collaboratively develop algorithms and code as well as analyse and visualise large data sets acquired from sources including satellite instruments, the International Space Station, and airborne and ground campaigns. The large data and high-performance computing required for MAAP, along with a shared code repository and catalogue, are stored and managed in the cloud. MAAP capabilities are supported and shared between NASA and ESA.

MAAP is said to provide access to NASA and ESA Earth science 'data' and is a model for open source science collaboration and analysis. It is the host platform for the first globally harmonised assessment of aboveground carbon – information that is vital for managing global climate change.

MAAP is the culmination of a two-year NASA and ESA effort. It is fully operational and will incrementally expand its user community over the coming months. The release of MAAP Version 2 in the spring of 2022 will add additional data sources, allowing scientists to tackle a broader range of Earth science questions.

The initial application of MAAP focuses on measurements of aboveground biomass as part of a global effort to determine the size and carbon content of Earth's forests. This data is critical for our understanding of climate change and forecasting its impact, and will be part of regular updates to the Intergovernmental Panel on Climate Change (IPCC).